

Five-mode versus general stationary fields: background caps

6 October 2026

Six joint two-window fits. Same pooled inverse-gain calibration, unchanged transmission matrix, uniform final-arc references, cyclic-offset correspondence and measured-total-normalized channel objective. No signal-relative weighting. One background field is shared across both windows; cone geometry, brightness and offset are window-specific.

Both families have a stationary coherent pupil field plus a noninterfering Stokes background. Five-mode uses the earlier two uniform-polarization and three dipole-derived vector modes (five complex coefficients); general uses ten complex transverse pupil modes spanning signal overlaps. Both are evaluated on the same optical quadrature and annular Fresnel/BFP model. Pair sums hold by construction. Different finite bases need not have literally nested background-power representations; comparison here is of the implemented physical families.

Background caps refer to the smaller of the two original, gain-corrected training maxima, so the shared background satisfies the stated cap in both windows. Broad cap retains the previous 95% of the stricter original minimum total intensity (about 56% of the smaller maximum). The actual fitted BG percentage is reported below, separately for each window.

$$J = \sum_{ij} (\varepsilon_{ij}/T_j^{train})^2 + 100 \sum_j \left[\max \left(0, \frac{\hat{r}_j - r_{max}}{1 - r_{max}} \right) \right]^6 + 10 \sum_{ij} [\min(0, \hat{S}_{ij})/T_j^{train}]^2.$$

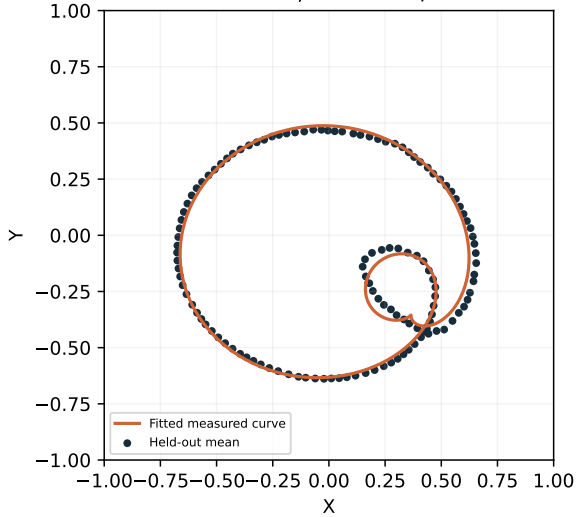
Here $\hat{S} = I^{train} - B - L(d^{fit})$. The radial term has a numerical denominator guard near zero pair intensity; raw corrected intensities are used for diagnostics. Penalties use training data only. Multiple starts and 8192-point refinement are used; convergence is reported, not a global-optimum guarantee.

Model/cap	Window	BG%	Meas. RMS	Signal RMS	Invalid	Conv.
5 / broad	30s	55.5	0.0064	0.0265	21	yes
5 / broad	10s	55.3	0.0104	0.0297	14	yes
5 / 10pct	30s	10.0	0.0249	0.0347	0	yes
5 / 10pct	10s	10.0	0.0272	0.0352	0	yes
5 / 5pct	30s	5.0	0.0327	0.0433	0	yes
5 / 5pct	10s	5.0	0.0347	0.0427	0	yes
10 / broad	30s	52.8	0.0049	0.0324	16	yes
10 / broad	10s	52.6	0.0097	0.0467	19	yes
10 / 10pct	30s	10.0	0.0114	0.0139	0	no
10 / 10pct	10s	10.0	0.0144	0.0175	0	no
10 / 5pct	30s	5.0	0.0196	0.0227	0	yes
10 / 5pct	10s	5.0	0.0231	0.0261	0	yes

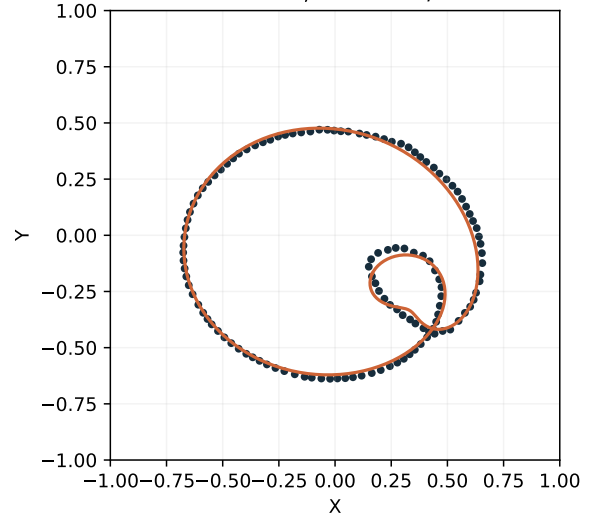
RMS values are four-channel residuals divided by measured total or fitted signal total respectively. Signal RMS is diagnostic only. Invalid counts include negative corrected channels or $r > r_{max}$. Signal-space curves show conditional fitted-interference subtraction, not self-consistent field inversion. Red triangles indicate points beyond fixed axes. A low cap may improve subtraction stability while worsening measured-space agreement; neither establishes correct rod angles. No commit or push.

30-31 s · measured space · fixed gains and steep penalty

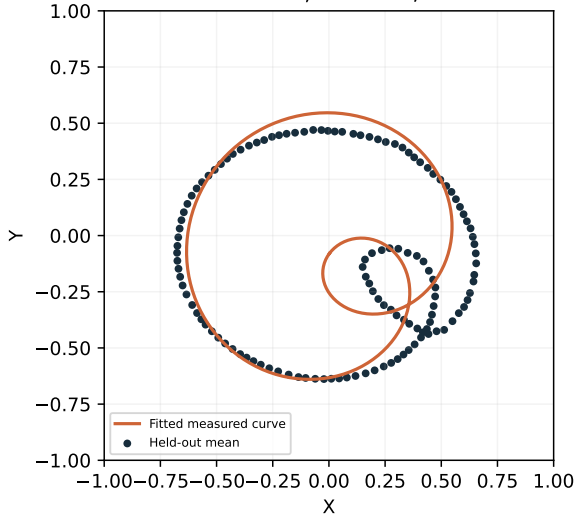
Five-mode · Broad cap
BG 55.5%, invalid 21/128



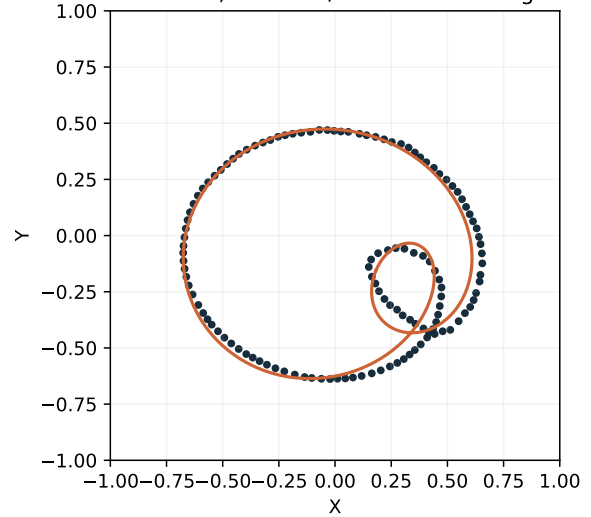
General stationary · Broad cap
BG 52.8%, invalid 16/128



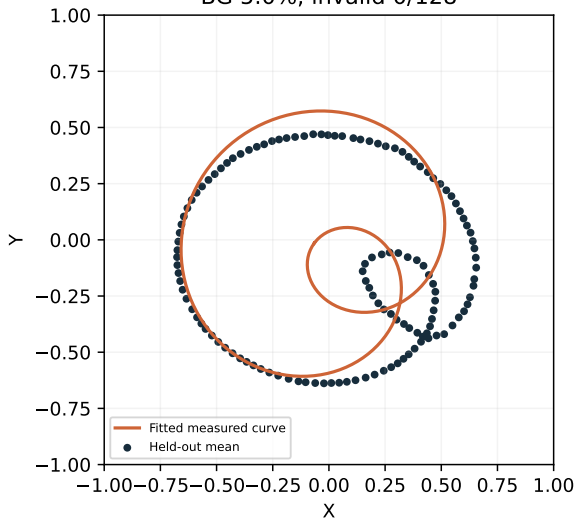
Five-mode · 10% cap
BG 10.0%, invalid 0/128



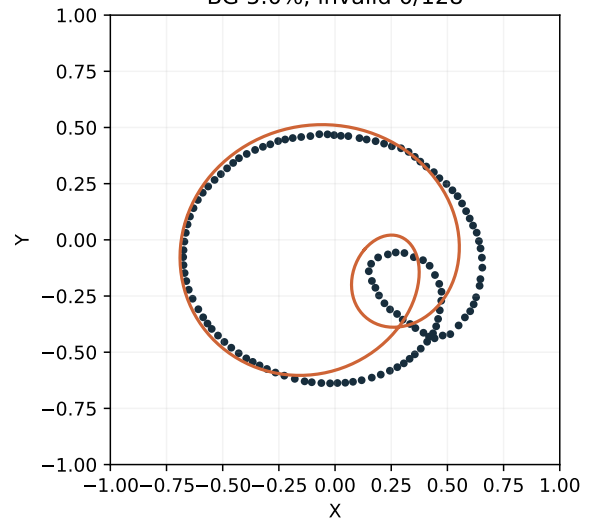
General stationary · 10% cap
BG 10.0%, invalid 0/128 · not converged



Five-mode · 5% cap
BG 5.0%, invalid 0/128

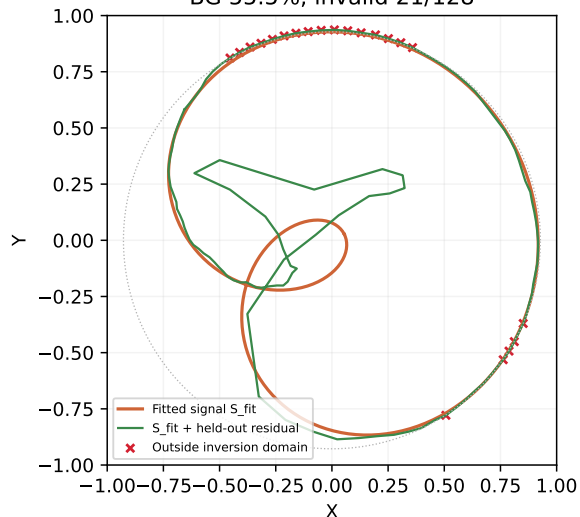


General stationary · 5% cap
BG 5.0%, invalid 0/128

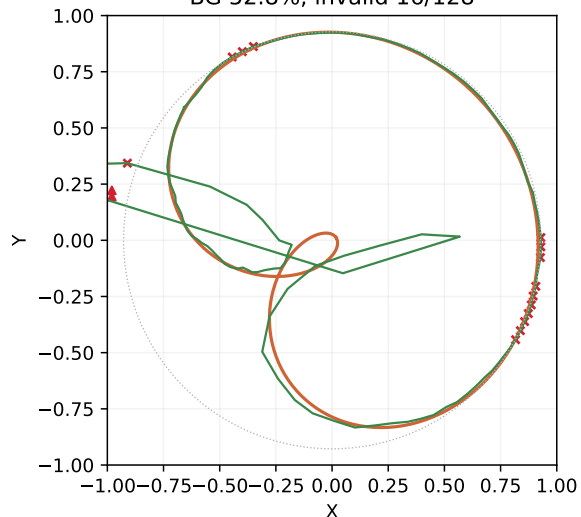


30-31 s · signal space · fixed gains and steep penalty

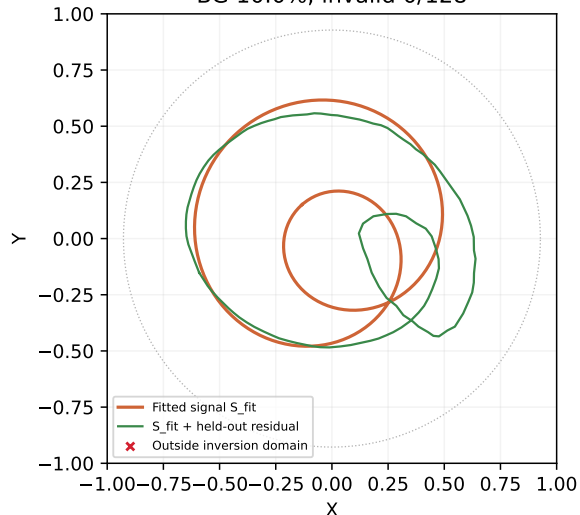
Five-mode · Broad cap
BG 55.5%, invalid 21/128



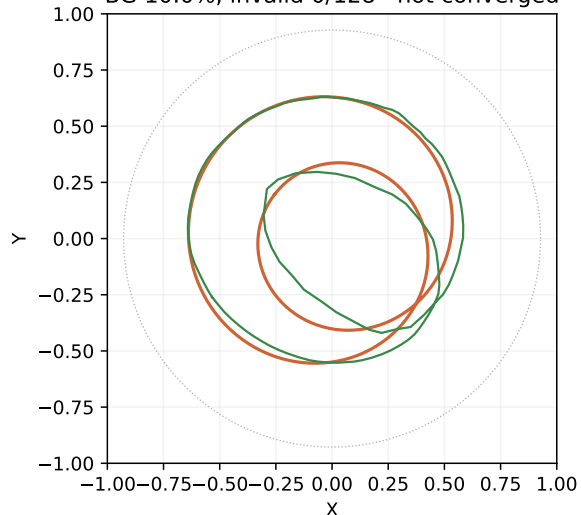
General stationary · Broad cap
BG 52.8%, invalid 16/128



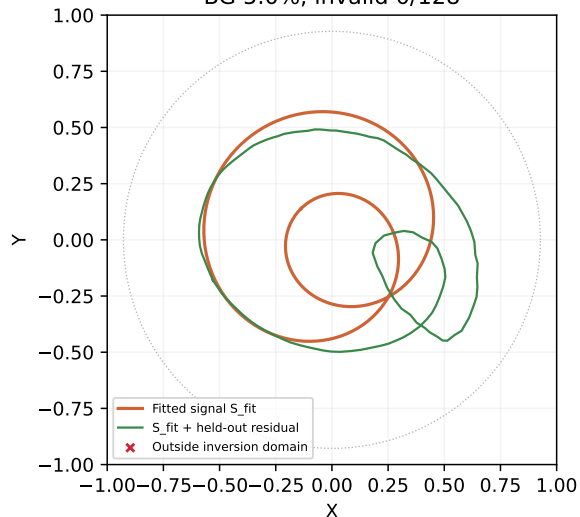
Five-mode · 10% cap
BG 10.0%, invalid 0/128



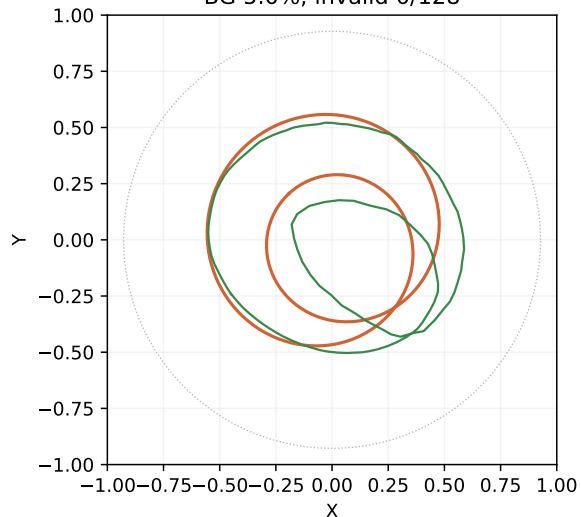
General stationary · 10% cap
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Five-mode · 5% cap
BG 5.0%, invalid 0/128

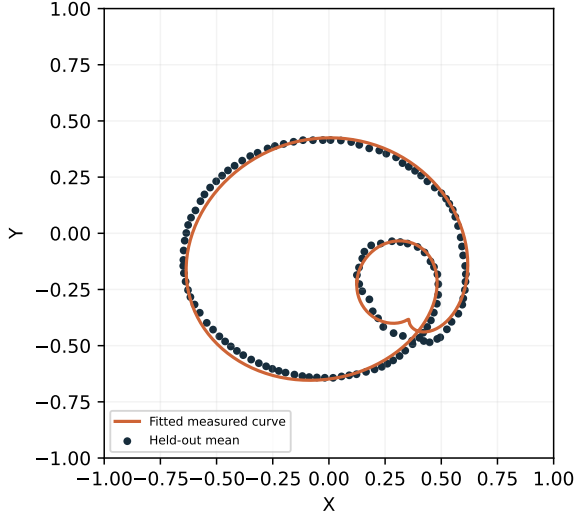


General stationary · 5% cap
BG 5.0%, invalid 0/128

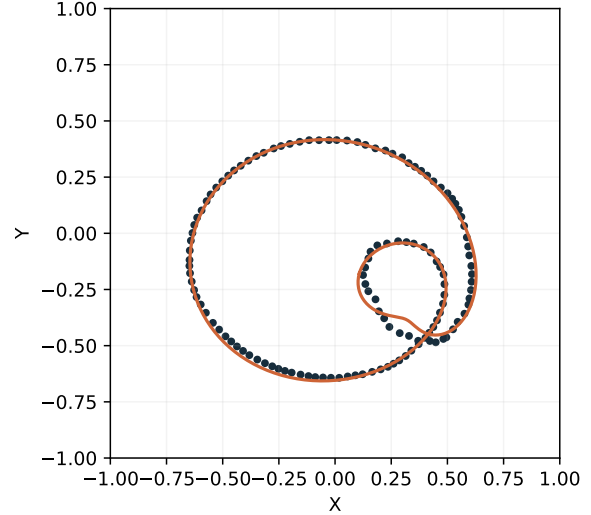


10-11 s · measured space · fixed gains and steep penalty

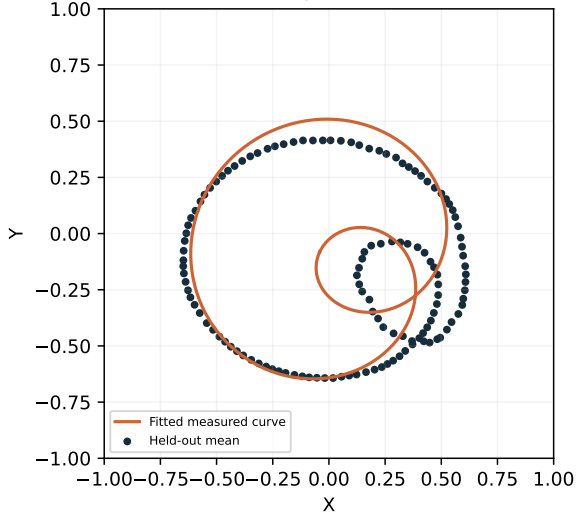
Five-mode · Broad cap
BG 55.3%, invalid 14/128



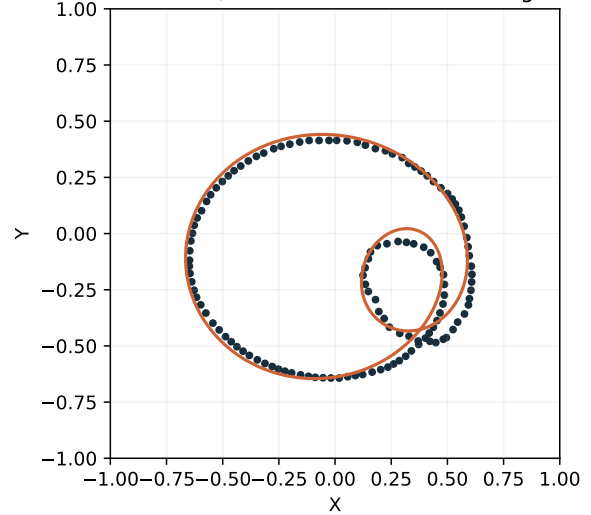
General stationary · Broad cap
BG 52.6%, invalid 19/128



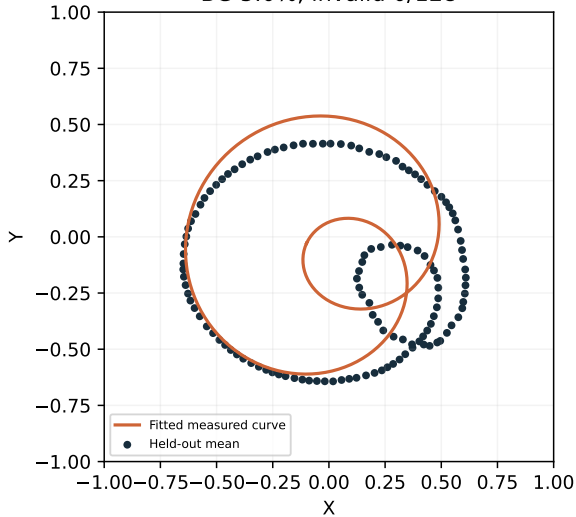
Five-mode · 10% cap
BG 10.0%, invalid 0/128



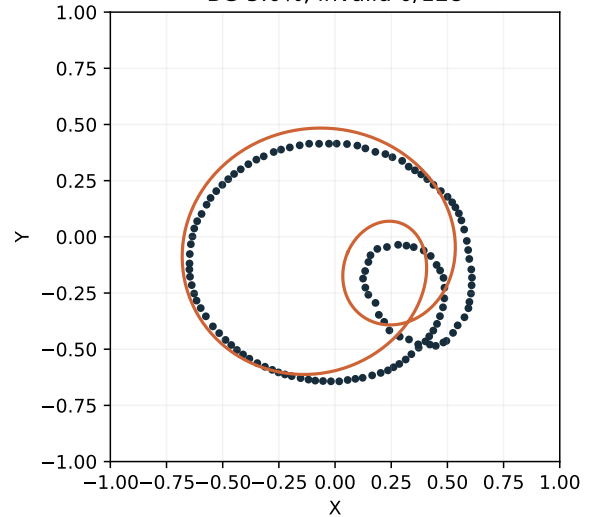
General stationary · 10% cap
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Five-mode · 5% cap
BG 5.0%, invalid 0/128

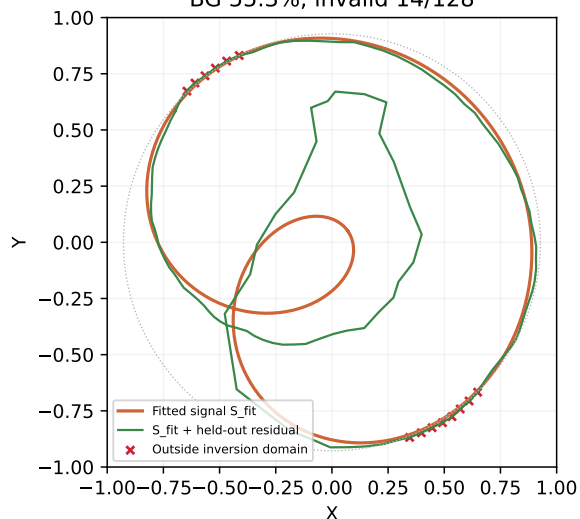


General stationary · 5% cap
BG 5.0%, invalid 0/128

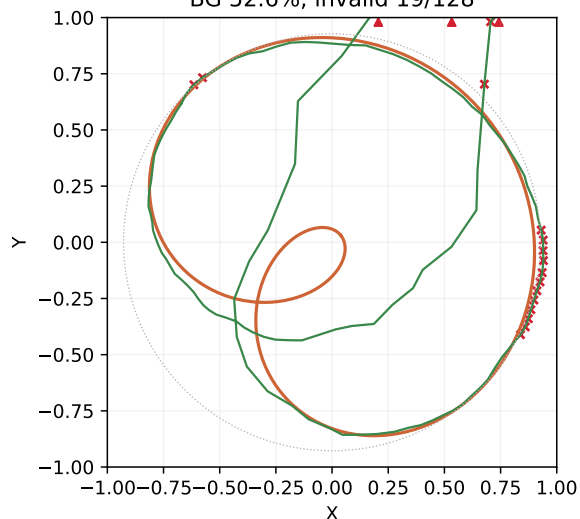


10-11 s · signal space · fixed gains and steep penalty

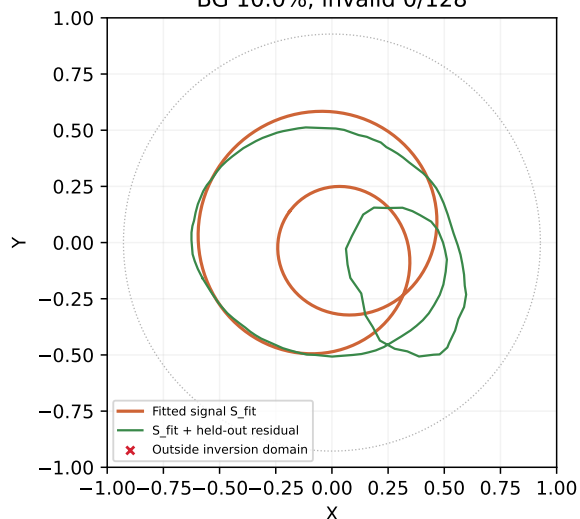
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BG 55.3%, invalid 14/128



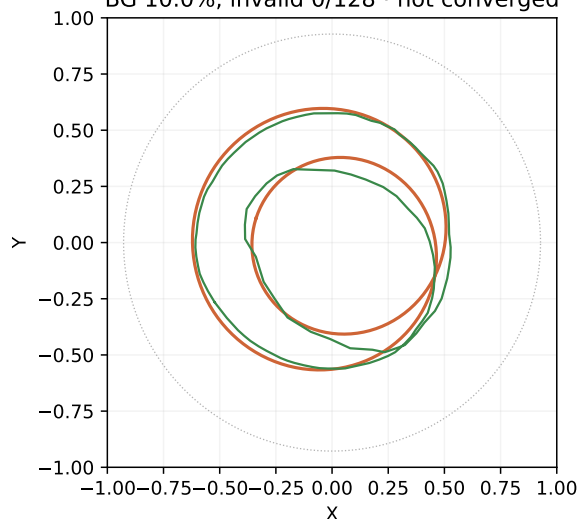
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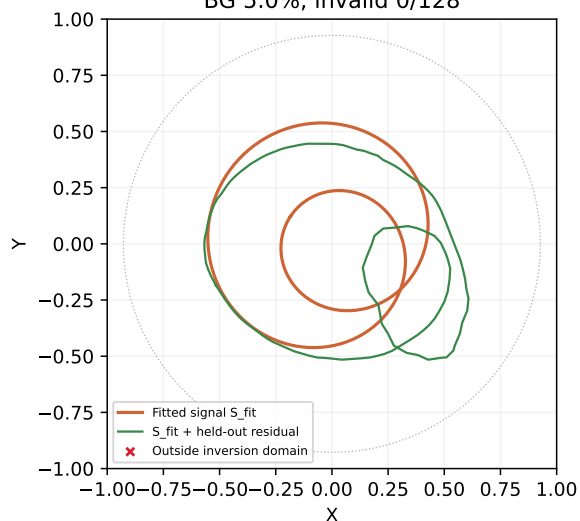
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Five-mode · 5% cap
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